

HCSM-00 (Ninth Revision)

Foundations, Notation, and Mapping

The Hodge Complex Standard Model:
Substrate, Hodge Complex, Spectral Midpoint, Processor,
Output Structures, Emergent Physics, and Cosmology

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, mixing matrices, spacetime dimension, graviton, Newton constant, and vacuum stability coefficient of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The substrate is selected by the unified substrate selection theorem. The emergent spacetime dimension is fixed by the topology of the Hodge-Dirac kernel. The Hodge complex of differential forms carries the operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation.

This paper establishes the framework's foundational objects and states the notation. All seven rank blockers of the framework's derivation program are closed. The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The paper establishes the following theorems. The 5-point Laplacian is the unique D_4 -invariant nearest-neighbour positive-semidefinite operator with kernel equal to the constants. The Hodge-Dirac operator satisfies $D^2|_{\Omega^k} = \Delta \cdot \text{id}$ on each form degree, with smallest nonzero eigenvalue $2 \sin(\pi/L)$ on Λ_L . The Lorentzian metric on $\ker D$ is the unique D_4 -invariant symmetric form that vanishes on pure-degree forms and has signature $(3, 1)$. The Kernel-Diagonal Theorem yields the EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ as three traces of one projector. The Λ -promotion mechanism is a pure projection onto the timelike direction of $\ker D$, and the refined dark energy density matches observation to 0.010%.

The charge sector is a rigidity-class result: five structural constraints reduce the charge assignment from $5^{14} \approx 6 \times 10^9$ configurations to exactly two, both with total charge $-1e$. The residual binary choice is resolved by the Midpoint Sorting Theorem of HCSM-23 Rev. 7.

The present revision incorporates four structural sharpenings. First, the mirror-cancellation theorem (HCSM-10) is integrated into the foundations: the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ pairs 50 of the 64 scalar modes into cancelling partners, and only the 14 self-paired modes at $\lambda = 4$ survive, giving the exact residual ratio $7/32$. Second, the $5 + 9$ split of the 14-mode multiplet (HCSM-11) is integrated: 5 Substrate Eigenmodes and 9 Mesh Resonances, with the base formula, shell values, and layer architecture now theorems rather than disclosed inputs. Third, the Hodge-HCSM

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interface (HCSM-12) is integrated: the natural state on the 56-dimensional Hodge sector is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, the processor channel is a completely positive trace-preserving map, and the entropy reduction is exactly $\Delta S = \log(7/5)$. Fourth, the identification of the A_1 isotypic block with physical color is now a theorem (HCSM-06, Theorem 5.4); the identification of the B_1 block with weak isospin is a labeling convention (HCSM-06, Theorem 5.10); and the hypercharge values and content structure are derived (HCSM-07, Theorems 8.1 and 8.3).

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1 Introduction

1.1 What HCSM is

The Hodge Complex Standard Model derives the Standard Model of particle physics from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The framework's central claim is that the Standard Model is not a fundamental description of nature but the residue left when a small, discrete, information-bearing lattice cancels against its own mirror.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

HCSM does not ask what fields exist or what symmetry groups are broken. It asks a single question: what substrate could produce exactly the Standard Model, and nothing else? The answer is a discrete lattice whose mirror cancellation leaves exactly the observed particle content. The mirror-cancellation theorem of HCSM-10 makes this statement precise: the Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, 50 of the 64 scalar modes pair into cancelling partners, and only the 14 self-paired modes at $\lambda = 4$ survive. The residual ratio is exactly $7/32$.

1.2 The single dimensionful anchor

The framework has exactly one dimensionful input: the Planck mass M_P . Every other dimensionful quantity is a derived multiple of an appropriate power of M_P . The dimensionless integers $d = 4$ and $N = 64$ are derived from the substrate (Section 2); the substrate size $L = 8$ is derived from the unified selection theorem (Section 2.6); and the emergent Lorentzian signature $(3, 1)$ is derived from the Hodge star on $\ker D$ (Section 3, Theorem 3.9).

At the particle level, the framework has zero empirical anchors. The identification of the 14 modes at $\lambda = 4$ with one Standard Model generation was until recently a disclosed input; it is now a theorem (HCSM-23 Rev. 7). The charge table was until recently a disclosed input up to a binary choice; it is now a theorem whose binary residue is resolved by the same sorting theorem.

As of the present revision, the gauge sector also has zero empirical anchors in the physical-identification step. The identification of the A_1 isotypic block with physical color $\mathfrak{su}(3)_c$ is a theorem (HCSM-06, Theorem 5.4): the A_1 block is the unique three-dimensional D_4 -invariant subspace of the 14-mode multiplet, and the dimension-and-invariance argument forces this identification. The identification of the B_1 block with physical weak isospin is a labeling convention (HCSM-06, Theorem 5.10), forced by the outer automorphism of D_4 .

1.3 Architectural levels

The framework's architecture is a chain from the substrate to observable physics, organized in seven levels.

Level	Name	Content
0	Substrate	8×8 torus, 5-point Laplacian, mirror cancellation, $50 + 14$ decomposition, residual $7/32$
1	Hodge complex	$\Omega^0 \oplus \Omega^1 \oplus \Omega^2$, $D = d + d^*$, Hodge-HCSM interface, entropic flatness
2	Spectral midpoint	$\lambda = 4$, 14 modes, D_4 orbits $(2, 8, 4)$
3	Processor	$T = I - P_C$, zero state s , five-layer loop, $5 + 9$ split, four-layer architecture
4	Output structures	Charge q_3 , mass map, $N/4$ labels, (S, D) pairs
5	Emergent physics	Gauge, flavor, Higgs, gravity, dark photon
6	Cosmology	kL crossover, dark energy, Λ -promotion

1.4 Continuum limit and emergent Lorentz invariance

The substrate is not a discretization of an underlying continuum. The framework's claim is the opposite: the continuum emerges from the substrate. The physical scale is set by the mass map and the kL crossover.

The 5-point Laplacian has the isotropic expansion

$$\lambda(n_1, n_2) = \frac{\pi^2}{16}(n_1^2 + n_2^2) + O(n^4). \quad (1)$$

2 Level 0: The Substrate

2.1 The periodic $L \times L$ torus

The substrate is the periodic square lattice

$$\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \quad (2)$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; equivalently, the coordinates x_1 and x_2 are each defined modulo L . Topologically, Λ_L is the torus $T^2 = S^1 \times S^1$.

The D_4 point group of the square acts on Λ_L by the eight rotations and reflections of the lattice. Its presentation is $D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle$.

2.2 The 5-point Laplacian is forced

Theorem 2.1 (Laplacian uniqueness). *Let $\Delta : \mathbb{C}^N \rightarrow \mathbb{C}^N$ be a linear operator on the substrate $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ satisfying:*

- (1) *D_4 -invariance: $[R, \Delta] = 0$ for every $R \in D_4$;*
- (2) *nearest-neighbour locality: $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbours of x ;*
- (3) *positivity and kernel: Δ is positive-semidefinite and $\ker \Delta$ is exactly the set of constant functions.*

Then Δ is uniquely determined up to an overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x + e). \quad (3)$$

Proof. Condition (2) writes $\Delta f(x) = a_0 f(x) + \sum_{\mu} a_{\mu} f(x + \mu)$. The four nearest-neighbour directions $\pm e_1, \pm e_2$ lie in a single D_4 orbit. Hence condition (1) forces $a_{\mu} = a$ for all four directions. Condition (3) forces $\Delta 1 = 0$, so $a_0 + 4a = 0$, giving $a_0 = -4a$. Normalizing $a = 1$ gives the 5-point Laplacian. $\square$

Remark 2.2. The 5-point Laplacian annihilates the constants by construction. Its eigenvalue on the plane wave $\phi_{n_1, n_2}(x) = \exp(2\pi i(n_1 x_1 + n_2 x_2)/L)$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad (4)$$

with $n_1, n_2 \in \{0, 1, \dots, L-1\}$. The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$.

2.3 The framework relation $N = d \cdot 2^d$

The framework uses the relation

$$N = d \cdot 2^d, \quad (5)$$

with N the number of substrate sites and d the emergent spacetime dimension. At $d = 4$ this gives $N = 4 \cdot 2^4 = 64$. Since $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ requires $N = L^2$, this gives $L = \sqrt{64} = 8$.

At $d = 4$, $N = 64$ is a perfect square. The integer solutions of $d \cdot 2^d = L^2$ form the infinite family $d \in \{4k^2 : k \geq 1\}$, with $d = 16, 36, 64, \dots$ also giving perfect squares. The physical selection of $d = 4$ comes from the unified selection theorem of Section 2.6.

2.4 Two derivations of $L = 8$

The substrate size $L = 8$ is selected by two independent derivations.

First derivation: equal entropy spacing. On the $L \times L$ torus, the standing-wave modes fall into discrete Shannon entropy classes.

Theorem 2.3 (Equal entropy spacing selects $L = 8$). *Among all integers $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally spaced Shannon entropy classes is $L = 8$. At $L = 8$ the three classes are 6.0 bits, 5.5 bits, and 5.0 bits, with uniform spacing 0.5 bits.*

Second derivation: maximal winding cancellation. The winding generator generalizes to any even L as $H_{\text{wind}}(L) = i \sin(\pi M/L)$ on the self-paired mode set of Λ_L .

Theorem 2.4 (Maximal cancellation selects $L = 8$). *Let $\mathcal{G}_L$ be the parity block graph of $H_{\text{wind}}(L)$ on the five nodes $\{A_{\text{odd}}, A_{\text{even}}, B_{\text{odd}}, B_{\text{even}}, AB\}$. The number of block couplings that vanish identically is*

$$\# \text{zeros}(G_L) = \begin{cases} 4, & L \equiv 2 \pmod{4}, \\ 7, & L \equiv 0 \pmod{4}, L \geq 12, \\ 10, & L \in \{4, 8\}. \end{cases} \quad (6)$$

Among the two maximally-cancelling sizes, only $L = 8$ carries the 14-mode multiplet.

The two derivations are independent: the first uses only the torus Laplacian's entropy classes; the second uses only the winding generator's parity block structure.

2.5 The 14-mode count

Theorem 2.5 (Midpoint multiplicity). *On the $L \times L$ torus with L even, the multiplicity of the self-paired eigenvalue $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(L) = 2(L - 1). \quad (7)$$

At $L = 8$ this gives 14.

Proof. The self-paired eigenvalue is the fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$. Its eigenspace consists of the modes (n_1, n_2) satisfying $\cos(2\pi n_1/L) + \cos(2\pi n_2/L) = 0$. Using the sum-to-product identity $\cos a + \cos b = 2 \cos((a+b)/2) \cos((a-b)/2)$, this becomes $2 \cos(\pi(n_1 + n_2)/L) \cos(\pi(n_1 - n_2)/L) = 0$. The solutions are $n_1 + n_2 \equiv L/2 \pmod{L}$ or $n_1 - n_2 \equiv L/2 \pmod{L}$. Each congruence has L solutions; the two congruences intersect in exactly two points. By inclusion-exclusion, the total is $2L - 2 = 2(L - 1)$. $\square$

2.6 Unified substrate selection

The substrate size $L = 8$ and the emergent dimension $d = 4$ are selected by four independent derivations in the current framework: equal entropy spacing on the torus Laplacian; maximal winding cancellation of the winding generator; Lorentzian signature on $\ker D$; and the Betti numbers of T^2 . This subsection shows that they are corollaries of a single selection theorem.

2.6.1 The two framework identities

Identity 1 (site-count relation). The number of substrate sites N and the emergent dimension d satisfy

$$N = d \cdot 2^d. \quad (8)$$

Identity 2 (mode-content relation). The self-paired mode count of the torus Laplacian at the midpoint eigenvalue equals the total particle content of one Standard Model generation:

$$\text{mult}_{\lambda=4}(L) = |F| + |B_{\text{bos}}|. \quad (9)$$

The right-hand side is derived content: the fermion-sector dimension $|F| = d(d+1)/2$ (HCSM-13, HCSM-19) and the boson-sector dimension $|B_{\text{bos}}| = d$ (HCSM-13, HCSM-19). At $d = 4$ they give $|F| = 10$, $|B_{\text{bos}}| = 4$, and total content 14.

The left-hand side is HCSM-04's multiplicity theorem:

$$\text{mult}_{\lambda=4}(L) = 2(L - 1). \quad (10)$$

2.6.2 The selection theorem

Theorem 2.6 (Unified substrate selection). *The system*

$$L^2 = d \cdot 2^d, \quad 2(L - 1) = d(d + 3)/2, \quad (11)$$

has the unique even-integer solution $L = 8$, $d = 4$, $N = 64$.

Proof. By Identity 1, $L = \sqrt{d \cdot 2^d}$. For L to be an even integer, $d \cdot 2^d$ must be a perfect square. Since 2^d is a perfect square iff d is even, write $d = 2m$. Then $d \cdot 2^d = 2m \cdot 2^{2m}$, which is a perfect square iff $2m$ is a perfect square, i.e. $m = 2k^2$ for integer $k \geq 1$. Hence

$$d = 4k^2, \quad L = 2^{2k^2+1} \cdot k. \quad (12)$$

Substituting into Identity 2 gives

$$2^{2k^2+1} \cdot k - 1 = k^2(4k^2 + 3). \quad (13)$$

Case $k = 1$: Left side $2^3 \cdot 1 - 1 = 7$. Right side $1 \cdot 7 = 7$. Satisfied.

Case $k = 2$: Left side $2^9 \cdot 2 - 1 = 1023$. Right side $4 \cdot 19 = 76$. Left exceeds right.

Case $k \geq 3$: The left side grows as $k \cdot 2^{2k^2+1}$, super-exponentially in k^2 ; the right side grows as $4k^4 + 3k^2$, polynomially. The left strictly dominates for all $k \geq 2$. Hence $k = 1$ is the unique solution, giving $d = 4$, $L = 8$, $N = 64$. $\square$

Corollary 2.7 (Uniqueness). *The pair $(L, d) = (8, 4)$ is the unique even-integer pair satisfying the two framework identities simultaneously.*

2.6.3 The four prior derivations as corollaries

Corollary 2.8 (Entropy spacing). *At $L = 8$, the substrate $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ has exactly three equally spaced Shannon entropy classes: 6.0, 5.5, and 5.0 bits.*

Corollary 2.9 (Maximal winding cancellation). *At $L = 8$, the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ has cancellation depth 10 on the five-node parity block graph, with kernel dimension $d(d+1)/2 = 10$.*

Corollary 2.10 (Lorentzian signature). *At $L = 8$, the Hodge star on $\ker D = H^0 + H^1 + H^2$ has eigenvalues $\{+1, -1, +i, -i\}$, giving a metric of signature $(3, 1)$ on $\ker D$.*

Corollary 2.11 (Betti numbers). *At $L = 8$, the kernel dimension is $d = \dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$.*

2.6.4 What this changes

Item	Before	After
Derivations of $L = 8$	2 independent	1 theorem, 1 corollary
Derivations of $d = 4$	2 independent	1 theorem, 2 corollaries
Status of $L = 8$	4 consistent derivations	1 theorem, 3 checks
Status of $d = 4$	4 consistent derivations	1 theorem, 3 checks

After the unified selection, the framework has exactly one open item outside the particle sector: the continuum limit.

2.7 The single lattice primitive

The mirror-cancellation theorem of HCSM-10 sharpens the substrate-level axiom structure. Where HCSM-01 lists three substrate-level disclosures, the mirror-cancellation theorem reduces these to a single primitive.

Definition 2.12 (Lattice primitive). *The single primitive of the HCSM substrate is: there exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.*

Every object in the framework, including the substrate size $L = 8$, the 5-point Laplacian, the Hodge complex, the mirror symmetry, the 14-mode multiplet, and the residual $7/32$, is a theorem of Definition 2.12. No physical input, no empirical anchor, and no fitted parameter is used at the substrate level.

2.8 Mirror cancellation

Definition 2.13 (Half-period translation). *The half-period translation is the map*

$$T_{\text{half}}(n_1, n_2) = ((n_1 + L/2) \bmod L, (n_2 + L/2) \bmod L). \quad (14)$$

Lemma 2.14. T_{half} is an involution on $\{0, \dots, L-1\}^2$.

Proof. $(n + L/2 + L/2) \bmod L = n \bmod L$ for both coordinates. $\square$

Lemma 2.15 (Cosine shift). *For $n \in \{0, \dots, L-1\}$ with L even,*

$$\cos\left(\frac{2\pi \cdot ((n + L/2) \bmod L)}{L}\right) = -\cos\left(\frac{2\pi n}{L}\right). \quad (15)$$

Proof. Let $m = (n + L/2) \bmod L$. If $n < L/2$, then $m = n + L/2$ and $\cos(2\pi m/L) = \cos(2\pi n/L + \pi) = -\cos(2\pi n/L)$. If $n \geq L/2$, then $m = n - L/2$ and $\cos(2\pi m/L) = \cos(2\pi n/L - \pi) = -\cos(2\pi n/L)$. Both cases coincide. $\square$

Theorem 2.16 (Mirror symmetry). *For every momentum pair (n_1, n_2) ,*

$$\lambda(T_{\text{half}}(n_1, n_2)) = 8 - \lambda(n_1, n_2). \quad (16)$$

Consequently, the spectrum of Δ on Λ_L is exactly mirror-symmetric: if λ is an eigenvalue with multiplicity m , then $8 - \lambda$ is also an eigenvalue with multiplicity m .

Proof. By (4), $\lambda(n_1, n_2) = 4 - 2\cos(2\pi n_1/L) - 2\cos(2\pi n_2/L)$. Applying T_{half} to both coordinates and using Lemma 2.15:

$$\begin{aligned} \lambda(T_{\text{half}}(n_1, n_2)) &= 4 - 2(-\cos(2\pi n_1/L)) - 2(-\cos(2\pi n_2/L)) \\ &= 4 + 2\cos(2\pi n_1/L) + 2\cos(2\pi n_2/L) \\ &= 8 - \lambda(n_1, n_2). \end{aligned} \quad (17)$$

T_{half} is a bijection by Lemma 2.14, so it pairs eigenvalues λ and $8 - \lambda$ at equal multiplicity. $\square$

Corollary 2.17. *Every eigenvalue $\lambda \neq 4$ appears in a mirror pair $(\lambda, 8 - \lambda)$ of equal multiplicity.*

2.9 The 50 + 14 decomposition and the residual 7/32

Theorem 2.18 (Self-paired eigenvalue). *The unique self-paired eigenvalue is $\lambda = 4$.*

Proof. $\lambda = 8 - \lambda$ implies $2\lambda = 8$, hence $\lambda = 4$. $\square$

Corollary 2.19 (50 + 14 decomposition). *At $L = 8$, the 64 modes of Λ_8 decompose as*

$$50 \text{ pair modes} + 14 \text{ residual modes at } \lambda = 4. \quad (18)$$

Proof. Immediate from Corollary 2.17 and Theorem 2.5: $(64 - 14)/2 = 25$ pairs, hence 50 pair modes, plus 14 residual modes. $\square$

Theorem 2.20 (General cancellation). *Let $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $F(\lambda) + F(8 - \lambda) = c$ for all λ and some constant $c \in \mathbb{R}$. Then*

$$\sum_m F(\lambda_m) = c \cdot \frac{L^2 - 2(L-1)}{2} + F(4) \cdot 2(L-1). \quad (19)$$

Proof. Group the L^2 modes into $(L^2 - 2(L - 1))$ paired modes and $2(L - 1)$ residual modes at $\lambda = 4$. Each pair $(\lambda, 8 - \lambda)$ with multiplicity m_p contributes $m_p \cdot c$ by hypothesis. Summing over all pairs gives $c \cdot (L^2 - 2(L - 1))/2$. Adding the residual contribution $2(L - 1) \cdot F(4)$ gives the result. $\square$

Corollary 2.21 ($L = 8$). *At $L = 8$,*

$$\sum_m F(\lambda_m) = 25c + 14F(4). \quad (20)$$

Proof. Substitute $L = 8$: $(64 - 14)/2 = 25$; $2 \cdot 7 = 14$. $\square$

Theorem 2.22 (Residual ratio $7/32$). *For the zero-point convention $F(\lambda) = \lambda/2$ and $L = 8$:*

$$\sum_m F(\lambda_m) = 128, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28, \quad \frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}, \quad (21)$$

where $\mathcal{R}$ is the residual set of 14 modes at $\lambda = 4$.

Proof. Compute $\sum_m \lambda_m$ exactly. Using the trigonometric identity $\sum_{n=0}^7 \cos(\pi n/4) = 0$, we have

$$\begin{aligned} \sum_m \lambda_m &= \sum_{n_1, n_2=0}^7 \left[4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) \right] \\ &= 4 \cdot 64 - 2 \cdot 8 \sum_{n=0}^7 \cos\left(\frac{\pi n}{4}\right) - 2 \cdot 8 \sum_{n=0}^7 \cos\left(\frac{\pi n}{4}\right) \\ &= 256 - 0 - 0 = 256. \end{aligned} \quad (22)$$

Therefore $\sum_m \lambda_m/2 = 128$. The residual contribution is $\sum_{m \in \mathcal{R}} \lambda_m/2 = 14 \cdot (4/2) = 28$. Ratio: $28/128 = 7/32$. $\square$

Theorem 2.23 (Convention independence). *Let $F(\lambda) = \alpha\lambda + \beta$ with $\alpha > 0$ and $F(4) \neq 0$. Then*

$$\frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}. \quad (23)$$

If $F(4) = 0$ (equivalently $\beta = -4\alpha$), both numerator and denominator vanish, and the ratio is undefined.

Proof. Compute:

$$\begin{aligned} \sum_{m \in \mathcal{R}} F(\lambda_m) &= 14F(4) = 14(4\alpha + \beta), \\ \sum_m F(\lambda_m) &= \alpha \sum_m \lambda_m + 64\beta = 256\alpha + 64\beta = 64(4\alpha + \beta). \end{aligned} \quad (24)$$

Ratio $= 14(4\alpha + \beta)/[64(4\alpha + \beta)] = 14/64 = 7/32$, provided $4\alpha + \beta \neq 0$. If $4\alpha + \beta = 0$, then $F(4) = 0$ and both sums vanish; the ratio is undefined. $\square$

Corollary 2.24 (Geometric content). *The convention-independent content of the cancellation is*

$$\frac{\sum_{m \in \mathcal{R}} \lambda_m}{\sum_m \lambda_m} = \frac{56}{256} = \frac{7}{32}. \quad (25)$$

The numerator $14 \cdot 4 = 56$ is multiplicity times the midpoint eigenvalue; the denominator $64 \cdot 4 = 256$ is total mode count times the midpoint eigenvalue.

2.10 The TF operator and the $|\Delta n|$ grading

Definition 2.25 (Diagonal flip). *The diagonal flip is $F : (n_1, n_2) \mapsto (n_2, n_1)$ on Λ_L .*

Definition 2.26 (TF operator). *The TF operator is the composition*

$$\text{TF} = T_{\text{half}} \circ F, \quad \text{TF}(n_1, n_2) = ((n_2 + L/2) \bmod L, (n_1 + L/2) \bmod L). \quad (26)$$

Lemma 2.27. $\text{TF}^2 = \text{id}$.

Proof. $\text{TF}^2(n_1, n_2) = \text{TF}(n_2 + L/2, n_1 + L/2) = (n_1 + L, n_2 + L) = (n_1, n_2) \bmod L$. $\square$

Theorem 2.28 (TF grading by $|\Delta n|$). *Let $\Delta n = n_2 - n_1$ on Λ_8 . On the 14 modes at $\lambda = 4$:*

- **TF-fixed modes:** *all modes with $\Delta n \equiv 4 \pmod{8}$. There are 8 such modes: $\{\nu, e, d, b, \mu, c, Z, H\}$.*
- **TF 2-cycle modes:** *all modes with $\Delta n \not\equiv 4 \pmod{8}$. There are 6 such modes, forming three 2-cycles: $\{u, \tau\}$, $\{s, \text{dark}\}$, $\{W, t\}$.*

Proof. TF-fixed iff $(n_2 + 4) \bmod 8 = n_1$ and $(n_1 + 4) \bmod 8 = n_2$, equivalent to $n_2 - n_1 \equiv 4 \pmod{8}$. By the parity constraint $n_1 \equiv n_2 \pmod{2}$ (which holds on every mode at $\lambda = 4$), the condition $\Delta n \equiv 4 \pmod{8}$ is equivalent to $|\Delta n| = 4$. Direct enumeration of the 14 modes confirms the stated classification. $\square$

Corollary 2.29 (TF-fixed $\Leftrightarrow$ higher $N/4$). *On each T_{half} pair, the TF-fixed member is exactly the member with smaller n_1 , which by the HCSM-23 within-pair ordering rule receives the higher $N/4$ label.*

Remark 2.30. Corollary 2.29 derives the HCSM-23 within-pair ordering rule from the TF operator of the substrate. What was stated as a rule in HCSM-23 is now a theorem of Definition 2.12.

2.11 Klein four-orbit structure

Definition 2.31 (Substrate involutions). *On Λ_8 , three commuting involutions act on the mode labels:*

$$\begin{aligned} F &: (n_1, n_2) \mapsto (n_2, n_1), \\ \text{TF} &: (n_1, n_2) \mapsto ((n_2 + 4) \bmod 8, (n_1 + 4) \bmod 8), \\ F \cdot \text{TF} &: (n_1, n_2) \mapsto ((n_1 + 4) \bmod 8, (n_2 + 4) \bmod 8). \end{aligned} \quad (27)$$

All three commute pairwise.

Theorem 2.32 (Klein four-orbit). *The four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful action of the Klein four-group $V_4 = \{\text{id}, F, \text{TF}, F \cdot \text{TF}\}$:*

$$\begin{aligned} F &: u \leftrightarrow s, \quad \text{dark} \leftrightarrow \tau, \\ \text{TF} &: u \leftrightarrow \tau, \quad s \leftrightarrow \text{dark}, \\ F \cdot \text{TF} &: u \leftrightarrow \text{dark}, \quad s \leftrightarrow \tau. \end{aligned}$$

The action is faithful, transitive, and free (no stabilizer).

Proof. Direct computation on the four modes. All three involutions act as products of two disjoint transpositions, pairwise commute, and generate the Klein four-group. No mode is fixed by any non-identity element, so the action is free. $\square$

Corollary 2.33. *The Klein four-orbit $\{u, s, \text{dark}, \tau\}$ is generated by the substrate involutions F and TF . This is the HCSM-27 Klein orbit, now derived from Definition 2.12.*

3 Level 1: The Hodge Complex

3.1 The three form degrees

On the 2-torus Λ_L , the Hodge complex of differential forms has three graded pieces:

$$\Omega^0 = \mathbb{R}^N, \quad \Omega^1 = \mathbb{R}^{2N}, \quad \Omega^2 = \mathbb{R}^N, \quad (28)$$

with total dimension $N + 2N + N = 4N$. At $N = 64$ this gives 256.

3.2 The exterior derivative and the codifferential

Let $e_1 = (1, 0)$ and $e_2 = (0, 1)$ be the two oriented lattice directions. The exterior derivative acts on 0-forms by

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \quad \mu \in \{1, 2\}, \quad (29)$$

and on 1-forms by

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x). \quad (30)$$

3.3 The Hodge-Dirac operator

Definition 3.1 (Hodge-Dirac operator). *The Hodge-Dirac operator is $D = d + d^*$, acting on the full Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$. In block form,*

$$D = \begin{pmatrix} 0 & d_0^* & 0 \\ d_0 & 0 & d_1^* \\ 0 & d_1 & 0 \end{pmatrix}, \quad (31)$$

a symmetric 256×256 real operator.

Theorem 3.2 (Hodge-Dirac square). *On the discrete torus Λ , the identity $D^2|_{\Omega^k} = \Delta \cdot \text{id}_{\Omega^k}$ holds exactly on each form degree $k = 0, 1, 2$.*

Proof. On Ω^0 , $D^2 = d_0^* d_0 = \Delta$. On Ω^1 , $D^2 = d_0 d_0^* + d_1^* d_1 = \Delta$ by direct component expansion. On Ω^2 , $D^2 = d_1 d_1^* = \Delta$. Cross terms are removed by nilpotency $d_1 d_0 = 0$. All three identities are verified at machine precision. $\square$

Remark 3.3 (No species doubling). The Hodge-Dirac operator on the discrete torus has no species doubling: its square equals the scalar Laplacian on each form degree. The low-energy spectrum is bounded away from zero by $2 \sin(\pi/L)$ except for the four zero modes.

3.4 The $\mathbb{Z}_2$ grading and the supercharge structure

Definition 3.4 (Parity operator). *The parity operator $(-1)^F$ on the Hodge complex is $+1$ on $\Omega^0 \oplus \Omega^2$ and -1 on Ω^1 . It $\mathbb{Z}_2$ -grades the complex into a bosonic sector $\Omega^0 \oplus \Omega^2$ and a fermionic sector Ω^1 .*

Proposition 3.5. $D(-1)^F = -(-1)^F D$.

Theorem 3.6 (Supercharge structure). *In the $\mathbb{Z}_2$ grading, the Hodge-Dirac operator decomposes as*

$$D = Q + Q^\dagger, \quad (32)$$

where Q is the off-diagonal block of D mapping the bosonic sector to the fermionic sector, and $Q^\dagger$ is its adjoint. The supercharges satisfy

$$Q^2 = (Q^\dagger)^2 = 0, \quad \{Q, Q^\dagger\} = D^2 = \Delta, \quad (33)$$

and the Hamiltonian $H = D^2/2 = \Delta/2$ satisfies $\{Q, Q^\dagger\} = 2H$. This is the standard $N = 1$ supersymmetric quantum mechanics algebra.

Proof. In the $\mathbb{Z}_2$ grading, D is off-diagonal:

$$D = \begin{pmatrix} 0 & Q^\dagger \\ Q & 0 \end{pmatrix}, \quad (34)$$

with $Q : \Omega^0 \oplus \Omega^2 \rightarrow \Omega^1$ and $Q^\dagger : \Omega^1 \rightarrow \Omega^0 \oplus \Omega^2$. Then

$$D^2 = \begin{pmatrix} Q^\dagger Q & 0 \\ 0 & Q Q^\dagger \end{pmatrix}. \quad (35)$$

The algebra follows from nilpotency and Theorem 3.2. $\square$

3.5 The kernel and the emergent dimension

Theorem 3.7 (Kernel dimension). *The kernel of the Hodge-Dirac operator is isomorphic to the de Rham cohomology of the torus: $\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R})$, with*

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (36)$$

Corollary 3.8 (Emergent dimension). $d = \dim \ker D = 4$.

Theorem 3.9 (Lorentzian signature). *The Hodge star $\star$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the ± 1 pair from the $(H^0 + H^2)$ sector, the $\pm i$ pair from the H^1 sector. The intersection form $Q(\alpha, \beta) = \int_\Lambda \alpha \wedge \beta$ on the even-degree sector and the Hodge inner product $\langle \alpha, \beta \rangle = \int_\Lambda \alpha \wedge \star \beta$ on the odd-degree sector combine into a metric*

$$g = Q|_{H^0+H^2} + \langle \cdot, \cdot \rangle|_{H^1} \quad (37)$$

of signature $(3, 1)$ on $\ker D$.

3.6 The spectral gap and absence of species doubling

Theorem 3.10 (Smallest nonzero eigenvalue of D). *On $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L even, the smallest nonzero eigenvalue of the Hodge-Dirac operator is exactly $\mu_{\min}(L) = 2 \sin(\pi/L)$.*

Corollary 3.11. *For every finite even L , the Hodge-Dirac operator has exactly four zero eigenvalues and the next eigenvalue is bounded away from zero by $2 \sin(\pi/L)$.*

3.7 The Hodge-HCSM interface: entropic flatness

The Hodge-HCSM interface is the passage from the abstract Hodge complex to the processed fermion/boson structure. The mirror-cancellation theorem of Section 2.8 provides the projection onto the 56-dimensional sector V_{56} , and the processor channel of this section provides the map onto the fermion sector.

3.7.1 The 56-dimensional Hodge sector

The Hodge complex at the self-paired eigenvalue $\lambda = 4$ is

$$V_{56} = \Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}, \quad (38)$$

of total dimension $14 + 28 + 14 = 56$. Its D_4 isotypic decomposition is

$$V_{56} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E, \quad (39)$$

with dimension check $9 + 5 + 7 + 7 + 28 = 56$.

3.7.2 The processor channel

By HCSM-16 (Definition 1, Lemma 2), the block-diagonal extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ is a Lie algebra homomorphism. The 56-dimensional processor is

$$T_{56} = \text{extend}(T) = T \oplus (T \otimes \mathbb{I}_2) \oplus T, \quad (40)$$

with $\text{rank}(T_{56}) = 10 + 20 + 10 = 40$ and $\dim \ker(T_{56}) = 16$.

Definition 3.12 (Processor channel). *The processor channel is the completely positive trace-preserving (CPTP) map*

$$\Phi_T(\rho) = T_{56} \rho T_{56} + \text{Tr}((\mathbb{I}_{56} - T_{56})\rho) |0\rangle\langle 0|_{\text{bos}}, \quad (41)$$

with $|0\rangle_{\text{bos}}$ a fixed reference state on the 16-dimensional bosonic kernel.

3.7.3 Entropic flatness and the entropy reduction

Theorem 3.13 (Entropic flatness). *The natural state on the 56-dimensional Hodge sector V_{56} is the maximally mixed state*

$$\rho_{\text{in}} = \frac{\mathbb{I}_{56}}{56}, \quad (42)$$

obtained as the image of the maximally mixed state on the full Hodge complex $\rho_{\text{full}} = \mathbb{I}_{256}/256$ under the mirror-cancellation projection. Its von Neumann entropy is

$$S_{\text{in}} = -\text{Tr}(\rho_{\text{in}} \log \rho_{\text{in}}) = \log 56 \approx 4.0253516907 \text{ nats}. \quad (43)$$

Theorem 3.14 (Processor channel output). *The processor channel Φ_T maps the maximally mixed input $\rho_{\text{in}} = \mathbb{I}_{56}/56$ to the maximally mixed state on the 40-dimensional image of T_{56} , with von Neumann entropy*

$$S_{\text{out}} = \log 40 \approx 3.6888794541 \text{ nats}. \quad (44)$$

Corollary 3.15 (Entropy reduction). *The entropy reduction effected by the processor is*

$$\Delta S = S_{\text{in}} - S_{\text{out}} = \log 56 - \log 40 = \log\left(\frac{7}{5}\right) \approx 0.3364722366 \text{ nats}. \quad (45)$$

Corollary 3.16 (Framework-constant form). *The entropy reduction has the clean HCSM-native form*

$$\Delta S = \log\left(\frac{\dim V_{56}}{\text{rank}(T_{56})}\right) = \log\left(\frac{56}{40}\right) = \log\left(\frac{7}{5}\right). \quad (46)$$

Corollary 3.17 (Landauer saturation). *Conditional on the HCSM measurement-irreversibility principle (HCSM-21, Theorem 12.1), the entropy reduction $\Delta S = \log(7/5)$ equals the exact deterministic Landauer bound for the processor's information-processing step.*

4 Level 2: The Spectral Midpoint

4.1 The self-paired eigenvalue

The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the scalar Laplacian spectrum has a unique fixed point at $\lambda = 4$. The eigenspace at $\lambda = 4$ has multiplicity $2(L - 1) = 14$ at $L = 8$. The 14 modes are

$$\begin{aligned}\mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}.\end{aligned}\tag{47}$$

4.2 The D_4 orbit decomposition

Proposition 4.1 (Orbit stabilizers). *Under the D_4 point group, the 14 modes decompose into the three orbits above of sizes $(2, 8, 4)$. The stabilizer orders of the three orbits are 4, 1, 2 respectively, satisfying the orbit-stabilizer identity $|\mathcal{O}_k| = |D_4|/|\text{Stab}(\mathcal{O}_k)| = 8/|\text{Stab}(\mathcal{O}_k)|$.*

4.3 The 56-dimensional Hodge lift

Theorem 4.2 (Fourteen-fold lift). *The 14-fold degeneracy of the scalar Laplacian at $\lambda = 4$ lifts to the Hodge complex as $14\Omega^0 + 28\Omega^1 + 14\Omega^2 = 56$.*

4.4 The D_4 decomposition of the 56-dimensional sector

Theorem 4.3 (D_4 decomposition). *As D_4 representations,*

$$\begin{aligned}\Omega^0|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E, \\ \Omega^1|_{\lambda=4} &= \Omega^0|_{\lambda=4} \otimes E = 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E, \\ \Omega^2|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E.\end{aligned}\tag{48}$$

The full 56-dimensional sector decomposes as $56 = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$.

5 Level 3: The Processor and Its Dynamics

5.1 The winding generator

Definition 5.1 (Winding matrix and winding generator). *Let $n_1, n_2 \in \mathbb{R}^{14}$ be the coordinate vectors of the 14 modes. The winding matrix is the rank-2 antisymmetric bilinear $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$. The winding generator is $H_{\text{wind}} = i \sin(\pi M/8)$, with $\sin$ applied elementwise.*

Theorem 5.2 (Characteristic polynomial). *The characteristic polynomial of H_{wind} is*

$$p(\lambda) = \det(\lambda \mathbb{I}_{14} - H_{\text{wind}}) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32).\tag{49}$$

Its spectrum is $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$.

5.2 The winding classes and the processor

Definition 5.3 (Winding class partition). *The 14 modes partition into three winding classes:*

$$\begin{aligned} A &= \{\nu, e, t\}, & |A| &= 3; \\ B_{\text{wind}} &= \{u, d, s, \mu, c, \tau, b\}, & |B_{\text{wind}}| &= 7; \\ C &= \{\text{dark}, W, Z, H\}, & |C| &= 4. \end{aligned} \quad (50)$$

The partition is the joint eigenspace decomposition of the n_1 -parity D and the processor T .

Definition 5.4 (Processor). *The processor T is the rank-10 orthogonal projection $T = I - P_C$ with P_C the diagonal projector onto winding class C .*

Proposition 5.5 (Fermion/boson split). *$T = I - P_C = P_A + P_{B_{\text{wind}}}$, with $\text{im}(T) = F = A \cup B_{\text{wind}}$ and $\ker(T) = C = B_{\text{bos}}$. The fermion and boson sector dimensions are $|F| = d(d+1)/2 = 10$ and $|B_{\text{bos}}| = d = 4$ (HCSM-13, HCSM-19). The internal ratios are $(|F| - |B_{\text{bos}}|)/|B_{\text{bos}}| = (d-1)/2 = 3/2$ and $|F|/|B_{\text{bos}}| = (d+1)/2 = 5/2$.*

5.3 The Kernel-Diagonal Theorem

Theorem 5.6 (Kernel-Diagonal). *Let $P_{\ker}$ be the orthogonal projector onto $\ker(H_{\text{wind}})$. The diagonal of $P_{\ker}$ in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\ker})_i = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (51)$$

Corollary 5.7 (Three coefficients from one diagonal). *With $P_{\ker}$ as above,*

$$\begin{aligned} c &= (d+1)/\text{Tr}(P_{\ker}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = 5/9, \\ \text{diag}(P_{\ker})|_{\mathcal{O}_1 \cup \mathcal{O}_2} / \text{diag}(P_{\ker})|_{\mathcal{O}_0} &= (d-1)/2 = 3/2, \\ \dim \ker(H_{\text{wind}}) &= 1 + (d-1)^2 = 10. \end{aligned} \quad (52)$$

5.4 The five-layer loop

The five-layer loop acts diagonally on the 14-mode multiplet:

$$M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}. \quad (53)$$

Theorem 5.8 (Loop-pass duration). *The loop-pass duration is selected by the deterministic Landauer bound: $T_{\text{pass}} = T_{\text{DME}} = 1/(2\gamma) = 0.7114000035169658 \dots$*

Theorem 5.9 (Exact ratio). $|M_A|/|M_B| = 3 = d - 1$.

5.5 The zero state

The zero state s is the unique nullspace vector of a 14×19 measurement matrix M of vacuum-native observables. The matrix has rank 13, nullity 1.

Theorem 5.10 (Zero-state structure). *The zero state satisfies $s(rs(m)) = -s(m)$ for all modes m , where rs is the diagonal reflection $(n_1, n_2) \mapsto (n_2, n_1)$. The two rs -fixed modes $W = (2, 2)$ and $t = (6, 6)$ have $s(W) = s(t) = 0$ exactly. The six independent amplitudes satisfy $a_3 = 2a_4$ and $a_5 = (2 - \sqrt{2})a_4$. Weight fractions: 3.3% on $\mathcal{O}_0$, 86.5% on $\mathcal{O}_1$, 10.2% on $\mathcal{O}_2$.*

Remark 5.11 (Zero state as sort key). The zero state is not only the unique nullvector of the measurement matrix; it is also the sort key of the charge assignment and one of the three native operators used in the Midpoint Sorting Theorem (HCSM-23 Rev. 7).

5.6 The 5 + 9 split

The 5 + 9 split of the 14-mode multiplet (HCSM-11) refines the processor's action. The classification is defined by the residual

$$\delta = N/4 - \text{base}, \quad (54)$$

where base is a reference depth in the entropy well.

Definition 5.12 (Substrate Eigenmode and Mesh Resonance). *A mode is a Substrate Eigenmode if $\delta = 0$; it is a Mesh Resonance if $\delta \neq 0$.*

Theorem 5.13 (Derived base formula). *For each mode at $\lambda = 4$, let (k, node) be its T_{half} pair label. The residual $\delta = N/4 - \text{base}$ is given by*

$$\begin{aligned} \delta_{\text{high}} &= (d-1)(2-k) + (d-1)^3 \cdot \mathcal{K}[\text{node} = \text{AB}], \\ \delta_{\text{low}} &= \text{frac}(N/4_{\text{low}}) + 2(d-1) \cdot \mathcal{K}[\text{node} = \text{AB}], \end{aligned} \quad (55)$$

where $\text{frac}(x) = x - \lfloor x \rfloor$ and $\mathcal{K}[\cdot]$ is the indicator function. The base is $\text{base} = N/4 - \delta$.

Theorem 5.14 (5 + 9 split). *Of the 14 modes at $\lambda = 4$, exactly 5 are Substrate Eigenmodes and exactly 9 are Mesh Resonances. The 5 Substrate Eigenmodes are the dark photon, the τ lepton, the W boson, the Z boson, and the Higgs boson. The 9 Mesh Resonances are the remaining 9 fermions.*

Theorem 5.15 (Shell formula). *The shell of each mode is determined by the $N/4$ value through the step function*

$$\text{shell}(N/4) = \begin{cases} 0, & N/4 \geq 50, \\ 1, & 18 \leq N/4 < 50, \\ 1.5, & 12 \leq N/4 < 18, \\ 2, & 6 \leq N/4 < 12, \\ 3, & N/4 < 6. \end{cases} \quad (56)$$

The thresholds are framework constants: $\{2(d-1), d(d-1), 2(d-1)^2, N-2(L-1)\} = \{6, 12, 18, 50\}$.

Theorem 5.16 (W/Z base formula). *The base values of the W and Z bosons are $\text{base}(W) = \text{base}(Z) = d/2 = 2$.*

5.7 The four-layer architecture

The 14-mode multiplet carries four independent structural layers (HCSM-11, Theorem 9.1).

Theorem 5.17 (Layer independence). *The 14-mode multiplet carries four independent structural layers:*

1. **Winding layer:** the kernel projector P_{ker} of H_{wind} , producing $c = 5/9$, $(d-1)/2 = 3/2$, and kernel dimension 10 (HCSM-14.a).
2. **Hartree layer:** the mode-density kernel K , producing the eigenvalue $7/32$ and the coefficient $26 = 2^{d+1} - (|F| - |B_{\text{bos}}|)$.
3. **Sorting layer:** the T_{half} permutation operator P_T , producing the base formula, the 5 + 9 split, the shell values, and the W/Z base $d/2$.

4. **Processing layer:** the shape parameters r_a and the coefficients $\{7.88, 4.23\}$ of $m_f(r)$, which are orthogonal to layers 1, 2, and 3.

Layers 1, 2, and 3 are HCSM-native and mutually compatible: $[P_T, K] = 0$, $[P_{\text{ker}}, F] = 0$, $[P_{\text{ker}}, D] = 0$, and K commutes with F and D . Layer 4 is orthogonal to all three.

Remark 5.18 (Structural meaning). Layer 4 is the framework's empirical anchor in the mass sector. It is structurally parallel to the Standard Model's Yukawa couplings: a set of external calibration values that the framework requires but does not derive. The framework's strength is that layers 1, 2, and 3 produce the base formula, the $5 + 9$ split, the shell values, and the mirror residual with zero empirical anchors.

5.8 The mirror residual as a Hartree eigenvalue

Theorem 5.19 (Mirror residual equals Hartree eigenvalue). *The top eigenvalue of the Hartree kernel K equals the mirror-cancellation residual of Section 2.9:*

$$\lambda_{\max}(K) = \frac{7}{32} = \frac{2d-1}{2^{d+1}}. \quad (57)$$

The corresponding eigenvector is the uniform superposition of all 14 modes.

Theorem 5.20 (T_{half} commutes with the Hartree kernel). *The T_{half} permutation operator P_T and the Hartree kernel K commute: $[P_T, K] = 0$.*

Theorem 5.21 (Coefficient 26). *The HCSM-30 processing function coefficient 26 satisfies*

$$26 = 2^{d+1} - (|F| - |B_{\text{bos}}|) = 32 - 6. \quad (58)$$

6 Level 4: Output Structures

6.1 The charge functional

The electric charge in units of $e/3$ is given by the functional

$$q_3 = F(n_1, \sigma), \quad \sigma = \text{sign}(s), \quad (59)$$

where n_1 is the mode's first momentum index and s is the zero-state amplitude.

Theorem 6.1 (Charge bijection). *The 14 modes have 14 pairwise distinct (n_1, σ) tuples. The map F is well-defined on the realized pairs and is a bijection onto the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$.*

Theorem 6.2 (σ -line sums). *The charge sums over the three σ -lines are*

$$\sum_{\sigma=-1} q_3 = 0, \quad \sum_{\sigma=0} q_3 = +5, \quad \sum_{\sigma=+1} q_3 = -8. \quad (60)$$

The total vacuum charge is $\sum_i q_{3i} = -3 = -1e$.

6.2 The rigidity class of the charge assignment

Theorem 6.3 (Charge rigidity class). *Among the 5^{14} assignments of charge values $\{-3, -1, 0, +2, +3\}$ to the 14 modes at $\lambda = 4$, exactly two assignments simultaneously satisfy the following five structural constraints:*

- (C1) **SM multiset.** *The multiset of charges is $\{-3, -1, 3, 0, 4, +2, 3, +3, 1\}$.*
- (C2) **σ -line sums.** *The charge sums over the three zero-state sign lines are 0, $d + 1 = 5$, and $-2d = -8$.*
- (C3) **F -pair sum rules.** *Within each $|\Delta n|$ shell, the pair charge sum is rank- d (for $|\Delta n| = 4$) or $(d + 1) - d$ rank (for $|\Delta n| = 2$).*
- (C4) **$n_1 + n_2 = 4$ line.** *The five modes on the line $n_1 + n_2 = 4$ carry the five charge values exactly once each.*
- (C5) **Processor bookkeeping.** *The image-of- T charge sum is -2 and the kernel-of- T charge sum is $+1$.*

Proof. Exhaustive enumeration. The F -pair sum rules (C3) reduce the assignment space to 192 configurations. Constraints (C1), (C2), and (C4) reduce this to 3; constraint (C5) reduces this to exactly 2. $\square$

Theorem 6.4 (Charge assignment solutions). *The two rigid solutions are*

$$\begin{aligned} Q_3^{(1)} &= (-3, 0, -1, -1, 2, -3, 0, 2, -3, -1, 3, 0, 0, 2), \\ Q_3^{(2)} &= (0, -3, 2, -1, -1, -3, 0, 2, -3, -1, 3, 0, 0, 2). \end{aligned} \tag{61}$$

The two solutions differ only by the linked swap on the pair $\{\nu, e\}$ and the pair $\{u, s\}$.

Theorem 6.5 (Total charge rigidity). *Both solutions have total charge $\sum_i q_{3i} = -3 = -1e$. Consequently, every assignment satisfying the five structural constraints has total charge $-1e$:*

$$\sum_i q_{3i} = -(d - 1) = -1e. \tag{62}$$

The total vacuum charge is a rigidity-class invariant.

Corollary 6.6 (Empirical residue of the charge sector). *The charge assignment is determined up to exactly one binary choice: which member of the pair $\{\nu, e\}$ carries charge 0 and which carries -3 . The Standard Model content resolves this choice: the electron e carries $-1e$ and the neutrino ν carries 0. This binary choice is now derived, not disclosed: HCSM-23 Rev. 7 Corollary 4.4 shows that the within-pair ordering rule of the Midpoint Sorting Theorem forces $\nu = (0, 4)$ and $e = (4, 0)$.*

Remark 6.7 (Physical reading: what the machine must not cancel). The rigidity theorem states that the machine cannot cancel its residual charge. Any attempt to modify the charge assignment away from the two-element family violates one of the five structural constraints. The total charge $-1e$ is not a choice made by the machine; it is the unique value permitted by the machine's own structure.

6.3 The mass map

The mass map is $m_i = A_{\text{mass}} \exp(-B_{\text{mass}} N/4_i)$, with

$$A_{\text{mass}} = (2d - 1)N/2 = 224 \text{ GeV}, \quad B_{\text{mass}} = (d - 1)kL^2/(2N^2) = 0.5411967 \dots \quad (63)$$

The map reproduces the twelve charged fermion masses and the W , Z , H , and t masses at RMS 0.0449 dex across thirteen orders of magnitude in mass. The coefficient $(d - 1)/2$ in B_{mass} is derived in HCSM-19 by two independent routes (the processor F/B split and the DNLS Hartree orbit trace). The mass map itself is derived in HCSM-33.

6.4 The Midpoint Sorting Theorem (summary)

The identification of the 14 modes with one Standard Model generation is now a theorem (HCSM-23 Rev. 7). The sorting uses the T_{half} involution $(n_1, n_2) \mapsto (n_1 + 4, n_2 + 4)$, the pair index $k = \min(n_1)$, the parity-block node from the sign of the signed coordinate $\mathcal{V}$ of the smaller- n_1 member, the (S, D) table of HCSM-23 Rev. 6, and the within-pair ordering rule.

7 Level 5: Emergent Physics

7.1 Gauge algebra from the D_4 commutant

The gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ is derived as a Lie subalgebra of the D_4 commutant on the 14-mode multiplet (HCSM-06). The A_1 isotypic block is uniquely identified with physical color by a dimension-and-invariance theorem (HCSM-06, Theorem 5.4). The B_1 block is identified with physical weak isospin up to the outer-automorphism labeling convention (HCSM-06, Theorem 5.10).

7.2 Hypercharge quantization

The hypercharge functional is a specific element of the 15-dimensional sector-coarse space (HCSM-06, Theorem 6.3; HCSM-07). The A_1 coefficients are $(0, -1/6, -1/2)$ and the B_1 coefficients are $(0, +1/2, +1/2)$, uniquely determined by the Standard Model content and the sector assignment (HCSM-07, Theorems 5.1 and 6.1). The hypercharge values are derived from the content structure and the anomaly cancellation conditions (HCSM-07, Theorem 8.1). The content structure is derived from the 14-mode charges, the D_4 isotypic multiplicities, and the Hodge-lift chirality structure (HCSM-07, Theorem 8.3). All values lie in $(1/6)\mathbb{Z}$.

7.3 Flavor: the CKM matrix

The Cabibbo angle satisfies $\sin \theta_{12} = \pi/14 = 0.2243995 \dots$, matching observation to 0.044% (HCSM-34, Theorem 1). The winding amplitude $a = 3/5$ is topological: $a = (|F| - |B_{\text{bos}}|)/|F| = 6/10$. The Wolfenstein parameters are $\lambda = \pi/14$, $A = (d/(d+1))(1 + \delta_{\text{edge}})$, and $\sqrt{\rho^2 + \eta^2} = (d+1)/(d(d-1)) = 5/12$ (HCSM-34, Theorem 3). The full CKM hierarchy is derived in HCSM-34, Theorems 5 and 9.

7.4 Flavor: the PMNS matrix

The neutrino $N/4$ spectrum is $(60, 60, 58)$, derived from the four spin structures of the torus with Dirac indices $\pm 1/4$ (HCSM-36, Theorem 4.1). The PMNS magnitudes match at RMS 0.035 (HCSM-36, Theorem 6.1). The mixing angles are $\theta_{12} = 34.2^\circ$, $\theta_{23} = 47.9^\circ$, $\theta_{13} = 8.5^\circ$; the Dirac CP phase is $\delta_{CP} \approx 197^\circ$. The generation phase $\phi_{\text{gen}} = (3/5)\pi$ is exact (HCSM-36, Section 5).

7.5 Higgs sector

The top-Yukawa coefficient is $y_t^2 = 1 - (4/3)\delta_{\text{edge}}$ with $4/3 = |\mathcal{O}_2|/(d-1)$ (HCSM-40). The top mass is $m_t = 172.688398$ GeV (PDG 172.690 ± 0.30 GeV). The tree-level quartic coupling is $\lambda_{\text{tree}} = \ln(kL)/[(d-1)\pi^2]$, and the electroweak vacuum expectation value is $v_{\text{EW}} = M_P \exp(-kL_{\text{phys}})$.

7.6 Gravity

The graviton is the self-paired $B_1 \oplus B_2$ curvature mode of $\text{Sym}^2(\Omega^1|_{\lambda=4})$, consisting of $54B_1 \oplus 54B_2 = 108$ modes (HCSM-45, HCSM-46). It is exactly massless at all orders by D_4 Schur orthogonality (HCSM-46). The Newton constant is $G_N = (d+1)^2/(8\pi M_P^2) = 25/(8\pi M_P^2)$ (HCSM-48).

7.7 Absence of a light dark photon

Theorem 7.1 (Absence of a light dark photon). *The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly over all mass scales.*

7.8 Spin chirality $\mathbb{Z}_2$ grading

The 56-dimensional Hodge lift at $\lambda = 4$ carries three commuting $\mathbb{Z}_2$ operators: $\gamma_5 = P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$ (Hodge form-parity), χ_{spin} (spin-structure parity), and $F : (n_1, n_2) \mapsto (n_2, n_1)$ (orientation flip). All three square to I and pairwise commute, generating $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ (HCSM-42).

7.9 Strong CP from orientation $\mathbb{Z}_2$

Theorem 7.2 (Strong CP). $\theta_{\text{QCD}} = \theta_{\text{eff}} = 0$ exactly.

Proof. On the 14-mode multiplet at $\lambda = 4$, $[F, \Delta] = 0$ and $\{F, H_{\text{wind}}\} = 0$ exactly. The winding matrix M is antisymmetric under F ; the odd function $\sin(\pi M/8)$ preserves the antisymmetry. The physical identification $H_{\text{wind}} \leftrightarrow F_{\mu\nu} F^{\mu\nu}$ gives $\theta_{\text{eff}} = 0$ exactly (HCSM-43). $\square$

7.10 Proca sector and $\mathfrak{u}(1)_Y$ origin

The $\mathfrak{u}(1)_Y$ gauge symmetry is generated by the hypercharge functional $\hat{Y}$ on $\Omega^0|_{\lambda=4}$, with $[\hat{Y}, T] = 0$, $[\hat{Y}, F] = 0$, and eigenvalues in $(1/6)\mathbb{Z}$. The Proca Lagrangian is

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4}W_{\mu\nu}^a W^{a\mu\nu} - \frac{1}{4}B_{\mu\nu} B^{\mu\nu} + \frac{1}{2}m_W^2 W_\mu^+ W^{-\mu} + \frac{1}{2}m_Z^2 Z_\mu Z^\mu, \quad (64)$$

with $m_W = gv_{\text{EW}}/2$ and $m_Z = \sqrt{g^2 + g'^2}v_{\text{EW}}/2$. The mass ratio $m_W/m_Z = \cos\theta_W = 0.8769$ matches PDG 0.8814 at 0.5%. Tree-level $\rho = 1$ exactly (HCSM-44).

8 Level 6: Cosmology

8.1 The kL crossover

The bare warp factor is exact: $kL_{\text{bare}} = (d-1)N/(d+1) = 3 \cdot 64/5 = 192/5 = 38.4$. The physical warp factor is $kL_{\text{phys}} = 38.442527$, matching $\ln(M_P/v_{\text{EW}})$ to 0.004%. The correction is $kL_{\text{phys}} - kL_{\text{bare}} = ((|D_4| - 1)/2)\delta_{\text{edge}} = (7/2)\delta_{\text{edge}}$ (HCSM-51).

8.2 Dark energy and the DME entropy-well normalization

The bare dark energy density is $\rho_{DE}^{\text{bare}} = (|F|/|B_{\text{bos}}|)m_s^4 = ((d+1)/2)m_s^4 = 2.4809 \times 10^{-47}\text{GeV}^4$ matching observation to 0.76%. The refined formula is $\rho_{DE} = ((d+1)/2)(1+c\delta)m_s^4$ with $c = (d+1)/(d-1)^2 = 5/9$ and $\delta = (1-\gamma)^2/(2\pi)$, giving $\rho_{DE} = 2.5003 \times 10^{-47}\text{GeV}^4$, matching observation to 0.010% (HCSM-52, HCSM-53).

The DME entropy-well normalization is $S_0 = v_{\text{EW}}^2 \cdot (7/40) \cdot (1 + \delta_{\text{edge}}/6) = 10630.722\text{GeV}^2$ matching the implied value at relative residual 2.44×10^{-6} (HCSM-53).

8.3 Vacuum stability

The stability coefficient is $\gamma = c_2/I(2) = 0.702839467990\dots$, with $I(2) = 1 - \frac{1}{2}\text{AGM}(1, \sqrt{2})$, $\text{AGM}(1, \sqrt{2}) = 1/(\sqrt{2}\eta(i)^2)$, and $\eta(i) = \Gamma(1/4)/(2\pi^{3/4})$. The stability theorem states $S''(0)/S_0 = -2\gamma < 0$: the equilibrium $D = 4$ is a strict local maximum of the vacuum entropy (HCSM-54).

8.4 The Λ -promotion mechanism

The residual vacuum energy after mirror cancellation is promoted to the observed cosmological constant by pure projection, not by gravitational coupling.

The five-layer loop $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$ acts on the 14-mode multiplet. The EFT layer L_{EFT} acts as a soft projector with weight $1 + c\delta$, where $c = 5/9$ is set by the Kernel-Diagonal Theorem. The residual energy projects onto the timelike direction of $\ker D$, with unit projection weight. The hidden $u(1)$ factors contribute zero.

The mechanism is therefore a projection, not a coupling. There is no gravitational suppression factor $1/M_P^2$; the residual vacuum energy is already at the right scale because it is topological. Combined with the absence of a light dark photon, this confirms that no gauge mediator exists between the hidden and visible sectors (HCSM-56.a).

9 Falsifiable Predictions

1. **No light dark photon.** The kinetic mixing parameter satisfies $\epsilon \equiv 0$ exactly. Falsifiable by SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC at $\epsilon > 10^{-12}$.
2. **Strong CP.** $\theta_{\text{QCD}} = 0$ exactly. Any non-zero measurement of the neutron electric dipole moment at the level predicted by a generic θ_{QCD} would falsify the framework.
3. **ρ parameter.** Tree-level $\rho = m_W^2/(m_Z^2 \cos^2 \theta_W) = 1$ exactly. Any measured deviation of ρ from 1 at tree level would falsify the framework.
4. **Fermion content.** One Standard Model generation contains exactly 14 particles. A fourth generation would falsify the framework.
5. **Kernel structure.** The winding kernel has dimension 10 on Λ_8 and the kernel-diagonal values are $\{1/2, 3/4\}$. Any modification of the spectrum of H_{wind} that changes these values would falsify the framework.
6. **Dark energy.** The dark energy density is $\rho_{DE} = ((d+1)/2)(1+c\delta)m_\nu^4$, matching observation to 0.010%. Any future measurement of ρ_{DE} that deviates from this formula at more than 0.1% would falsify the framework.

7. **Charge rigidity.** The charge assignment on the 14-mode multiplet satisfies exactly two structurally consistent configurations, both with total charge $-1e$. A measurement of the vacuum's total electromagnetic charge that differs from $-1e$ (per one Standard Model generation) would falsify the rigidity theorem.
8. **Identification.** The Midpoint Sorting Theorem (HCSM-23 Rev. 7) assigns the 14 modes uniquely. A Standard Model particle with an $N/4$ label outside the set $\{60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5\}$ would falsify the framework.
9. **A_1 uniqueness.** The physical color algebra is the $\mathfrak{su}(3)$ subalgebra on the A_1 isotypic block, forced by the dimension-and-invariance theorem. A different embedding of $\mathfrak{su}(3)_c$ in the D_4 commutant would falsify Theorem 5.4 of HCSM-06.
10. **Hypercharge values.** The hypercharge values are uniquely determined by the content structure and anomaly cancellation. A different set of hypercharge values satisfying the same content structure and anomaly conditions would falsify Theorem 8.1 of HCSM-07.
11. **Mirror cancellation.** The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, and the residual ratio is exactly $7/32$. Any deviation from these exact structural identities would falsify the mirror-cancellation theorem (HCSM-10).
12. **TF grading and Klein orbit.** The TF operator grades the 14-mode multiplet by $|\Delta n|$, and the four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful Klein four-group action. Any modification of these exact structural identities would falsify HCSM-10.
13. **$5 + 9$ split.** The 14-mode multiplet splits into exactly 5 Substrate Eigenmodes and 9 Mesh Resonances. Any modification of the derived base formula or shell values would falsify HCSM-11.
14. **Layer independence.** The 14-mode multiplet carries four independent structural layers; layers 1, 2, 3 are HCSM-native and mutually compatible, while layer 4 requires external calibration. Any HCSM-native derivation of the processing layer shape parameters r_a would falsify the layer-independence theorem (HCSM-11).
15. **Entropic flatness.** The natural state on the 56-dimensional Hodge sector is the maximally mixed state $\rho_{\text{in}} = \mathbb{I}_{56}/56$, with entropy reduction $\Delta S = \log(7/5)$. Any deviation from these exact structural identities would falsify the Hodge-HCSM interface (HCSM-12).

10 Theorem Inventory

The framework's theorems are catalogued by their home paper. New theorems from the present revision cycle are marked with $\star$.

Theorem	Location
Laplacian uniqueness	HCSM-01 Thm 2.1
$d \cdot 2^d = 64, L = 8$	HCSM-01 Cor 4.1
Two derivations of $L = 8$	HCSM-01 §4, HCSM-14.b
$2(L - 1) = 14$	HCSM-01 Thm 4.4
Unified substrate selection	HCSM-00 §2.6
Single lattice primitive	$\star$ HCSM-00 §2.7

Theorem	Location
Mirror symmetry	★ HCSM-00 Thm 2.8
$50 + 14$ decomposition	★ HCSM-00 Cor 2.10
Residual ratio $7/32$	★ HCSM-00 Thm 2.12
Convention independence	★ HCSM-00 Thm 2.13
TF grading by $ \Delta n $	★ HCSM-00 Thm 2.16
TF-fixed iff higher $N/4$	★ HCSM-00 Cor 2.17
Klein four-orbit	★ HCSM-00 Thm 2.19
Hodge-Dirac square	HCSM-02 Thm 3.2
Supercharge algebra	HCSM-02 Thm 4.4
Kernel dimension $= 4$	HCSM-02 Thm 3.8
Spectral gap $2 \sin(\pi/L)$	HCSM-02 Thm 3.12
Lorentzian metric uniqueness	HCSM-03 Thm 3.10
Entropic flatness	★ HCSM-00 Thm 3.14
Processor channel output	★ HCSM-00 Thm 3.15
Entropy reduction $\log(7/5)$	★ HCSM-00 Cor 3.16
Landauer saturation	★ HCSM-00 Cor 3.17
Self-paired eigenvalue $\lambda = 4$	HCSM-04 Thm 3.2
D_4 orbit decomposition	HCSM-04 Thm 5.1
Support classes	HCSM-04 Prop 6.2
IPRs	HCSM-04 Prop 7.2
Entropy classes	HCSM-04 Cor 8.3
D_4 character	HCSM-05 Thm 4.1
Irreducible decomposition	HCSM-05 Thm 6.1
n_1 -parity orbit indicator	HCSM-05 Thm 7.3
Joint (D, T) class partition	HCSM-05 Thm 7.7
Sector-coarse gauge algebra	HCSM-06 Thm 5.10
A_1 uniqueness for physical color	HCSM-06 Thm 5.4
B_1/B_2 outer automorphism	HCSM-06 Thms 5.8, 5.10
Hypercharge coefficients	HCSM-07 Thms 5.1, 6.1
Hypercharge values from anomaly	HCSM-07 Thm 8.1
Content structure from HCSM data	HCSM-07 Thm 8.3
Orientation-flip $\mathbb{Z}_2$	HCSM-09
Winding class partition	HCSM-14 Thm 15
Kernel-Diagonal Theorem	HCSM-14.a Thm 1
Three coefficients from one diagonal	HCSM-14.a Cor 2
Maximal cancellation selects $L = 8$	HCSM-14.b Thm 4.4.1
Hartree orbit trace identity	HCSM-19 Prop 9.2
Loop-pass duration	HCSM-21 Thm 12.1
Exact ratio $ M_A / M_B = 3$	HCSM-20 Thm 9.4
Zero-state structure	HCSM-22 Thm 12
Midpoint Sorting Theorem	HCSM-23 Rev. 7 Thm 5.3
Charge bijection	HCSM-24 Thm 4.1
σ -line sums	HCSM-24 Thm 5.1
Non-existence of closed form	HCSM-24 Thm 9.1
Sorting-machine interpretation	HCSM-24 Thm 10.1
Framework-constant σ -line sums	HCSM-24 Thm 10.6

Theorem	Location
F -pair sum rules by shell	HCSM-24 Thm 10.7
Charge rigidity class	HCSM-24 Thm 11.1
Charge assignment solutions	HCSM-24 Thm 11.2
Total charge rigidity	HCSM-24 Thm 11.3
Empirical residue of the charge sector	HCSM-24 Cor 11.4
Derived base formula	★ HCSM-00 Thm 5.13
$5 + 9$ split	★ HCSM-00 Thm 5.14
Shell formula	★ HCSM-00 Thm 5.15
W/Z base formula	★ HCSM-00 Thm 5.16
Mirror residual as Hartree eigenvalue	★ HCSM-00 Thm 5.18
T_{half} commutes with Hartree	★ HCSM-00 Thm 5.19
Coefficient 26	★ HCSM-00 Thm 5.20
Layer independence	★ HCSM-00 Thm 5.21
Top-Yukawa coefficient $4/3$	HCSM-40
Absence of light dark photon	HCSM-41 Thm 5
Spin chirality $\mathbb{Z}_2$ grading	HCSM-42
Strong CP $\theta_{\text{QCD}} = 0$	HCSM-43 Thm 6
Proca sector	HCSM-44
Graviton masslessness by Schur	HCSM-45, 46
G_N normalization	HCSM-48
kL correction $(7/2)\delta_{\text{edge}}$	HCSM-51
EFT coefficient $c = 5/9$	HCSM-52
S_0 normalization	HCSM-53
Vacuum stability $\gamma = c_2/I(2)$	HCSM-54
Λ -promotion is a projection	HCSM-56.a

All seven rank blockers of the framework’s derivation program are closed. As of the present revision, the physical identification of the A_1 and B_1 blocks is derived (color) or a labeling convention (weak isospin); the hypercharge values and content structure are derived; the mirror cancellation, $5 + 9$ split, and Hodge-HCSM interface are integrated. The framework has zero empirical anchors in the particle, gauge, and hypercharge sectors.

11 Reproducibility

All numerical claims in the HCSM paper stack are verified at machine precision in IEEE-754 double precision. The verification scripts use only standard libraries (NumPy, SciPy, SymPy, mpmath) and construct the objects they need from HCSM-native definitions. Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the stack.

The complete HCSM paper stack, verification scripts, and source are available at

<https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM>

12 Conclusion

This paper has established the foundation of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem (Section 2.6). The substrate carries the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ with $D^2 = \Delta$ exactly on each form degree. The 14 modes at the self-paired eigenvalue $\lambda = 4$ are identified with one Standard Model generation. The processor T and its five-layer loop are the machine's operation. The output structures (charge, mass, flavor, gravity, cosmology) are the residue after mirror cancellation.

The framework's most consequential structural results are seven. First, the Kernel-Diagonal Theorem unifies the framework's three leading coefficients under a single projector: the EFT coefficient $c = 5/9$, the mass-map coefficient $(d-1)/2 = 3/2$, and the kernel dimension $10 = (d-1)^2 + 1$ are three traces of the diagonal of the projector onto $\ker(H_{\text{wind}})$. Second, the Λ -promotion mechanism is closed: the residual vacuum energy is promoted by pure projection onto the timelike direction of $\ker D$, and the hidden $\mathfrak{u}(1)$ VEVs vanish. Third, the framework's coverage of the Standard Model gauge content (absence of a light dark photon, spin-chirality $\mathbb{Z}_2$ grading, strong-CP solution, Proca sector, topological protection of ϕ_{gen}) is complete. Fourth, the charge sector is a rigidity-class result: five structural constraints reduce the charge assignment from 5^{14} configurations to exactly two, both with total charge $-1e$, and the residual binary choice is resolved by the Midpoint Sorting Theorem. Fifth, the identification thesis is closed: the 14 modes are uniquely sorted into the 14 Standard Model particles by native HCSM data, with zero empirical anchors remaining in the particle sector. Sixth, the mirror-cancellation theorem (HCSM-10) is integrated into the foundations: the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ pairs 50 of the 64 scalar modes into cancelling partners, only the 14 self-paired modes at $\lambda = 4$ survive, and the residual ratio is exactly $7/32$. Seventh, the $5 + 9$ split of the 14-mode multiplet (HCSM-11) and the Hodge-HCSM interface (HCSM-12) are integrated: 5 Substrate Eigenmodes and 9 Mesh Resonances; the natural state on the 56-dimensional Hodge sector is the maximally mixed state; the processor channel is a completely positive trace-preserving map; and the entropy reduction is exactly $\log(7/5)$.

The present revision closes the physical identification of the gauge and hypercharge sectors. The A_1 isotypic block is uniquely identified with physical color by a dimension-and-invariance theorem (HCSM-06, Theorem 5.4). The B_1 block is identified with physical weak isospin up to the outer-automorphism labeling convention (HCSM-06, Theorem 5.10). The Standard Model content structure and hypercharge values are derived from HCSM-native data and anomaly cancellation (HCSM-07, Theorems 8.1 and 8.3). The framework now has zero empirical anchors in the particle, gauge, and hypercharge sectors.

The framework has a single dimensionful anchor (the Planck mass M_P) and derives every other quantity from the substrate at $d = 4$ and $L = 8$.

The framework's claim is that the Standard Model is what the substrate looks like when you observe it.

Physics is what the machine cannot cancel. The Hodge complex is the machine. The $-1e$ is what the machine must not cancel.

References

- [1] S. Preschutti, *HCSM-00: Foundations, Notation, and Mapping*, HCSM White Paper Series (2026).
- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).

- [3] S. Preschutti, *HCSM-02: The Hodge Complex on Λ* , HCSM White Paper Series (2026).
- [4] S. Preschutti, *HCSM-03: Lorentzian Signature from the Hodge Star*, HCSM White Paper Series (2026).
- [5] S. Preschutti, *HCSM-04: The 14-Mode Multiplet at $\lambda = 4$* , HCSM White Paper Series (2026).
- [6] S. Preschutti, *HCSM-05: D_4 Representation Theory of the 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [7] S. Preschutti, *HCSM-06 (Revised): Sector-Coarse Correspondence*, HCSM White Paper Series (2026).
- [8] S. Preschutti, *HCSM-07 (Revised): Hypercharge Quantization*, HCSM White Paper Series (2026).
- [9] S. Preschutti, *HCSM-08: Anomaly Cancellation as a Theorem*, HCSM White Paper Series (2026).
- [10] S. Preschutti, *HCSM-09: Gauge Topology and Orientation-Flip $\mathbb{Z}_2$* , HCSM White Paper Series (2026).
- [11] S. Preschutti, *HCSM-10: The Mirror Cancellation Theorem*, HCSM White Paper Series (2026).
- [12] S. Preschutti, *HCSM-11 (Second Revision): The $5 + 9$ Split of the 14-Mode Multiplet*, HCSM White Paper Series (2026).
- [13] S. Preschutti, *HCSM-12 (Second Revision): The Hodge-HCSM Interface: Entropic Flatness*, HCSM White Paper Series (2026).
- [14] S. Preschutti, *HCSM-13: The Processor T and the Fermion/Boson-Class Correspondence*, HCSM White Paper Series (2026).
- [15] S. Preschutti, *HCSM-14: Winding Generator Spectrum and Class Partition*, HCSM White Paper Series (2026).
- [16] S. Preschutti, *HCSM-14.a: The Kernel-Diagonal Theorem*, HCSM White Paper Series (2026).
- [17] S. Preschutti, *HCSM-14.b: Maximal Cancellation Selects $L = 8$* , HCSM White Paper Series (2026).
- [18] S. Preschutti, *HCSM-16: Loop Operator on the Full 56-Dimensional Hodge Sector*, HCSM White Paper Series (2026).
- [19] S. Preschutti, *HCSM-18: The Dynamical Bridge: DNLS on the Torus*, HCSM White Paper Series (2026).
- [20] S. Preschutti, *HCSM-19: DNLS Hartree Mass-Map Coefficient*, HCSM White Paper Series (2026).
- [21] S. Preschutti, *HCSM-20: The Motion Theorem*, HCSM White Paper Series (2026).
- [22] S. Preschutti, *HCSM-21: The Time Theorem*, HCSM White Paper Series (2026).
- [23] S. Preschutti, *HCSM-22: The Zero State*, HCSM White Paper Series (2026).

- [24] S. Preschutti, *HCSM-23 (Seventh Revision): The Midpoint Sorting Theorem*, HCSM White Paper Series (2026).
- [25] S. Preschutti, *HCSM-24 (Revised): The Charge Theorem*, HCSM White Paper Series (2026).
- [26] S. Preschutti, *HCSM-27: Klein Orbit Structure*, HCSM White Paper Series (2026).
- [27] S. Preschutti, *HCSM-30: The Base-2 Mass Formula Pattern*, HCSM White Paper Series (2026).
- [28] S. Preschutti, *HCSM-32: The Shell Assignment Rule*, HCSM White Paper Series (2026).
- [29] S. Preschutti, *HCSM-33: Mass Map*, HCSM White Paper Series (2026).
- [30] S. Preschutti, *HCSM-34: The CKM Matrix*, HCSM White Paper Series (2026).
- [31] S. Preschutti, *HCSM-36: The PMNS Matrix*, HCSM White Paper Series (2026).
- [32] S. Preschutti, *HCSM-40: The Higgs Mechanism*, HCSM White Paper Series (2026).
- [33] S. Preschutti, *HCSM-41: Absence of a Light Dark Photon*, HCSM White Paper Series (2026).
- [34] S. Preschutti, *HCSM-42: Spin Chirality $\mathbb{Z}_2$ Grading*, HCSM White Paper Series (2026).
- [35] S. Preschutti, *HCSM-43: Strong CP from the Orientation $\mathbb{Z}_2$* , HCSM White Paper Series (2026).
- [36] S. Preschutti, *HCSM-44: The Proca Sector*, HCSM White Paper Series (2026).
- [37] S. Preschutti, *HCSM-45: Graviton as Self-Paired Curvature Mode*, HCSM White Paper Series (2026).
- [38] S. Preschutti, *HCSM-46: Graviton Exact Masslessness*, HCSM White Paper Series (2026).
- [39] S. Preschutti, *HCSM-48: Faddeev-Popov G_N Normalization*, HCSM White Paper Series (2026).
- [40] S. Preschutti, *HCSM-51: The kL Crossover*, HCSM White Paper Series (2026).
- [41] S. Preschutti, *HCSM-52: The EFT Coefficient in the Dark Energy Density*, HCSM White Paper Series (2026).
- [42] S. Preschutti, *HCSM-53: Dark Energy Density and the DME Entropy-Well Normalization*, HCSM White Paper Series (2026).
- [43] S. Preschutti, *HCSM-54: Vacuum Stability and the Stability Coefficient*, HCSM White Paper Series (2026).
- [44] S. Preschutti, *HCSM-56: Vacuum EFT*, HCSM White Paper Series (2026).
- [45] S. Preschutti, *HCSM-56.a: Λ -Promotion as Pure Projection*, HCSM White Paper Series (2026).
- [46] J. Dodziuk, *Finite-difference approach to the Hodge theory of harmonic forms*, Amer. J. Math. **98**, 79 (1976).
- [47] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., CRC Press (2003).
- [48] C. Voisin, *Hodge Theory and Complex Algebraic Geometry I*, Cambridge University Press (2002).
- [49] Particle Data Group, Phys. Rev. D **110**, 030001 (2024).

A Web Crawler Metadata

A.1 Dublin Core

DC.title	HCSM-00 (Ninth Revision): Foundations, Notation, and Mapping
DC.creator	Preschutti, Stanley
DC.creator.orcid	0009-0004-5445-1744
DC.date.issued	2026-09
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DC.type	Text / Preprint / White Paper
DC.format	application/pdf; text/x-latex
DC.language	en
DC.publisher	Entropia Research Institute
DC.rights	All rights reserved
DC.subject	Hodge Complex Standard Model; HCSM; foundations; notation; mapping; 8×8 torus; Hodge complex; spectral midpoint; processor; output structures; emergent physics; cosmology; Laplacian uniqueness; Hodge-Dirac square; Lorentzian signature; Kernel-Diagonal Theorem; charge rigidity class; total charge invariance; Midpoint Sorting Theorem; identification thesis; zero empirical anchors; A_1 uniqueness; B_1/B_2 outer automorphism; hypercharge values; content structure; Λ -promotion; mirror cancellation; $7/32$; TF grading; Klein four-orbit; $5 + 9$ split; layer independence; entropic flatness; processor channel
DC.description	This paper establishes the foundational objects of the Hodge Complex Standard Model and states the notation. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, selected by the unified substrate selection theorem. The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ on each form degree. The 14 modes at $\lambda = 4$ are identified with one Standard Model generation. The ninth revision integrates the mirror-cancellation theorem (HCSM-10), the $5 + 9$ split (HCSM-11), and the Hodge-HCSM interface (HCSM-12) into the foundations. The framework now has zero empirical anchors in the particle, gauge, and hypercharge sectors.
DC.identifier	HCSM-00-2026-09-Rev9

A.2 BibTeX

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@article{PreschuttiHCSM00Rev9,  
  author = {Preschutti, Stanley},  
  title = {{HCSM-00} (Ninth Revision): Foundations, Notation,
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        and Mapping},
journal = {HCSM White Paper Series},
year    = {2026},
month   = {September},
note    = {ORCID: 0009-0004-5445-1744},
keywords = {Hodge Complex Standard Model, HCSM, foundations,
            notation, mapping, 8x8 torus, Hodge complex,
            spectral midpoint, processor, output structures,
            emergent physics, cosmology, Laplacian uniqueness,
            Hodge-Dirac square, Lorentzian signature,
            Kernel-Diagonal Theorem, charge rigidity class,
            total charge invariance, Midpoint Sorting Theorem,
            identification thesis, zero empirical anchors,
            A1 uniqueness, B1/B2 outer automorphism,
            hypercharge values, content structure,
            Lambda-promotion, mirror cancellation, 7/32,
            TF grading, Klein four-orbit, 5+9 split,
            layer independence, entropic flatness,
            processor channel, falsifiable predictions},
abstract = {This paper establishes the foundational objects of
            the Hodge Complex Standard Model. The substrate is
            the periodic 8x8 torus  $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ , selected
            by the unified substrate selection theorem. The
            Hodge-Dirac operator  $D = d + d^*$  satisfies
             $D^2 = \Delta$  on each form degree. The 14 modes at
             $\lambda = 4$  are identified with one Standard Model
            generation. The ninth revision integrates the
            mirror-cancellation theorem (HCSM-10), the 5+9
            split (HCSM-11), and the Hodge-HCSM interface
            (HCSM-12) into the foundations. The framework now
            has zero empirical anchors in the particle, gauge,
            and hypercharge sectors.}
}

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